SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

MV Switchgear Panel 12kV (SWGR-MV-12kV) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 800 mm in front for withdrawal of the circuit breaker truck; 100 mm at the rear unless rear-access type
- Operating temperature: -5 to +40 °C
- Degree of protection: IP4X
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check interlocks (circuit breaker, earthing switch, doors)
2. Perform power-frequency withstand test of the main circuit
3. Check protection relay settings by secondary injection
4. Verify CT polarity and VT ratio
5. Rack in and test closing and opening

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Partial discharge alarm in switchgear	Condensation due to anti-condensation heater off	TG-SWGR-MV-12kV §3.1
Partial discharge at cable compartment	Cable termination workmanship defect	TG-SWGR-MV-12kV §3.2
Cable compartment door cannot be opened	Earthing switch not closed (interlock)	TG-SWGR-MV-12kV §3.3
Busbar joint overheating	Insufficient joint torque	TG-SWGR-MV-12kV §3.4

On-site commissioning support can be requested through the Global CS Center. For mv breakers & switchgear the request is handled by Field Service.